# Akoizo Original Practice - Form 2 - Answer key

### Division B | Disease Detectives | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

### ANSWER KEY - Review only after submitting.

**1. (2 points) What strengthened Snow's work beyond a map?**

Unit: History of epidemiology and causal reasoning.
Worked-case data: Illness is more common among water-source A users, who are also substantially older. What should be investigated?

A. Ignoring denominators

B. Assuming all deaths shared one source

C. Exposure-based population comparisons

D. Only a memorable story

**Correct: C**

**2. (2 points) Selecting 50 cases and 50 controls primarily defines…**

A. A complete cohort automatically

B. A time series only

C. A case-control design

D. A randomized trial

**Correct: C**

**3. (2 points) The null value for an ordinary RR is…**

Unit: Interpreting results and interrupting transmission.
Worked-case data: Observed risks are 0.4 and 0.1 in a closed cohort. Explain the measures and propose the next analytic step.

A. 100

B. -1

C. 1

D. 0

**Correct: C**

**4. (2 points) Which must hold for ordinary causation?**

Unit: History of epidemiology and causal reasoning.
Worked-case data: Illness is more common among water-source A users, who are also substantially older. What should be investigated?

A. The largest group must be exposed

B. The map must be colorful

C. Relevant exposure precedes outcome

D. Outcome always precedes exposure

**Correct: C**

**5. (2 points) A full banquet roster investigated afterward is…**

Unit: Identifying study designs.
Worked-case data: Investigators interview every attendee on a banquet roster after an outbreak and compare illness by food consumed. Identify the design.

A. An ecological study only

B. A randomized trial

C. A retrospective cohort

D. Necessarily case-control

**Correct: C**

**6. (2 points) 0.4 minus 0.1 is…**

A. 3 percentage points

B. A 0.3% difference

C. 30 percentage points

D. 30 times the risk

**Correct: C**

**7. (2 points) What can age differences produce?**

Unit: History of epidemiology and causal reasoning.
Worked-case data: Illness is more common among water-source A users, who are also substantially older. What should be investigated?

A. No possible effect on comparisons

B. A new case definition necessarily

C. Confounding in a suitable causal structure

D. Automatic proof of causation

**Correct: C**

**8. (2 points) Random assignment and random sampling are…**

Unit: Identifying study designs.
Worked-case data: Investigators interview every attendee on a banquet roster after an outbreak and compare illness by food consumed. Identify the design.

A. Both proof of no bias

B. Both required for every survey

C. Different procedures

D. Synonyms

**Correct: C**

**9. (2 points) Vaccination primarily acts on…**

Unit: Interpreting results and interrupting transmission.
Worked-case data: Observed risks are 0.4 and 0.1 in a closed cohort. Explain the measures and propose the next analytic step.

A. The map projection

B. All reporting delays

C. Host susceptibility for the relevant agent

D. Every environmental reservoir

**Correct: C**

**10. (4 points) Why is mapping alone insufficient?**

Unit: History of epidemiology and causal reasoning.
Worked-case data: Illness is more common among water-source A users, who are also substantially older. What should be investigated?

answer: It locates cases but may omit population denominators and alternative explanations. (4 points).

**11. (4 points) Why not compute risk from an ordinary case-control sample fraction?**

Unit: Identifying study designs.
Worked-case data: Investigators interview every attendee on a banquet roster after an outbreak and compare illness by food consumed. Identify the design.

answer: Investigators choose case/control numbers; those fractions do not represent source-population disease occurrence. (4 points).

**12. (4 points) Write an RR = 4 interpretation.**

answer: The exposed group had four times the observed risk of the defined outcome versus the reference during follow-up, subject to study validity. (4 points).

**13. (4 points) Explain a comparison group's purpose.**

Unit: History of epidemiology and causal reasoning.
Worked-case data: Illness is more common among water-source A users, who are also substantially older. What should be investigated?

answer: It estimates what outcomes might look like under a different exposure or intervention, subject to comparability assumptions. (4 points).

**14. (4 points) Name a recall-bias example.**

Unit: Identifying study designs.
Worked-case data: Investigators interview every attendee on a banquet roster after an outbreak and compare illness by food consumed. Identify the design.

answer: Cases may remember or report a suspected food more readily than controls after publicity. (4 points).

**15. (4 points) Why does significance not prove cause?**

Unit: Interpreting results and interrupting transmission.
Worked-case data: Observed risks are 0.4 and 0.1 in a closed cohort. Explain the measures and propose the next analytic step.

answer: Chance is only one explanation; bias, confounding, and temporal or mechanistic issues can remain. (4 points).

**16. (4 points) What is confirmation bias here?**

Unit: History of epidemiology and causal reasoning.
Worked-case data: Illness is more common among water-source A users, who are also substantially older. What should be investigated?

answer: Seeking or emphasizing only evidence supporting a preferred hypothesis while neglecting contradictory or discriminating observations. (4 points).

**17. (4 points) Explain ecological fallacy.**

Unit: Identifying study designs.
Worked-case data: Investigators interview every attendee on a banquet roster after an outbreak and compare illness by food consumed. Identify the design.

answer: A relationship among group averages need not describe the individual-level relationship. (4 points).

**18. (4 points) Name three chain links.**

Unit: Interpreting results and interrupting transmission.
Worked-case data: Observed risks are 0.4 and 0.1 in a closed cohort. Explain the measures and propose the next analytic step.

answer: Examples: agent, reservoir, portal of exit, transmission mode, portal of entry, susceptible host. (4 points).